

Computational Validation of Bayesian FVC Trajectory Models

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1 Scope and disposition

This report evaluates implementation checks, sampling diagnostics, and a single synthetic parameter-recovery experiment. It reads the recorded research and recovery outputs and an explicit QA receipt. The common base seed is 4224. It does not certify external predictive validity, clinical utility, causal interpretation, or general parameter-recovery calibration.

Disposition: computational verification complete. Residual checks additionally identify a substantive serial-dependence limitation, independently of computational acceptance.

2 Research configuration and sampling checks

The prespecified research analysis has 24 configurations: three main fits, 15 patient-fold fits, four prior-scale variants, and two duplicate-handling variants. There are 24 recorded diagnostic rows and 34 recorded sampling attempts (10 retries). Attempts are counted separately from configurations; an initial failed attempt is retained in the metadata.

Analysis	Fits	Max R-hat	Min bulk	Min tail	Div.	Depth	Min BFMI
Research	24	1.0096	481	900	0	0	0.551
Recovery	3	1.0044	922	1662	0	0	0.597

Gates are rank-normalized R-hat < 1.01 , bulk and tail ESS ≥ 400 , no post-warmup divergences, no tree-depth hits, and per-chain E-BFMI ≥ 0.3 . Research slope MCSE has recorded maximum 0.0154 mL/week. The report recomputes gates from the extrema rather than relying only on the saved pass flag. These numerical gates do not establish that the Gaussian residual model is adequate.

3 Synthetic parameter recovery

Estimand	Truth	Estimate	Absolute error	Tolerance
C overall slope	-4.9760	-5.2504	0.2744	1.5000

All quantities in this table are in mL/week. The acceptance rule is absolute error strictly below the recorded tolerance. The calculation is independently checked against the saved truth and estimate. This is one seeded synthetic dataset and one targeted slope-recovery check; it is not simulation-based calibration or a repeated-simulation coverage study.

4 Model-adequacy limitation

Model	Observed	Replicated 95% interval	Tail fraction	Flag
H	0.107	[-0.055, 0.050]	0.000	yes
C	0.110	[-0.054, 0.052]	0.000	yes

The statistic is pooled adjacent-visit residual correlation within patients ordered by time. Observed entries are medians of draw-specific discrepancies; tail fractions compare replicated and observed discrepancies within each posterior draw. Fractions below 0.025 or above 0.975 flag lack of agreement, not calibrated hypothesis-test significance. The flagged discrepancy indicates remaining serial dependence. Nominal population credible intervals and new-patient predictive intervals retain a questionable residual-independence assumption, so calibration and predictive adequacy may be affected.

5 Evidence receipt and reproducibility

Required evidence	Result
Verified research manifest and seed	passed
All 24 research configurations and diagnostic gates	passed
Attempt counts agree with the manifest	passed
Research artifact hashes match	passed
Synthetic recovery manifest and seed	passed
All three recovery models meet diagnostic gates	passed
Recovery error is below its recorded tolerance	passed
Recovery artifact hashes match	passed
QA receipt matches both analysis hashes	passed
Unit and numerical tests	passed
Compiled synthetic smoke workflow	passed
Clean-directory synthetic reproduction	passed

Software checks are asserted only when the supplied receipt contains literal true values and its analysis hashes match both run manifests. Missing, false, malformed, or mismatched evidence cannot establish a pass. Research diagnostic, fit-metadata, and posterior-check hashes and recovery diagnostic and recovery-summary hashes are compared with the recorded artifact hashes. The receipt’s local path and host details are not published.

The receipt covers unit and numerical checks, a compiled synthetic smoke workflow, and reproduction from a clean directory. These checks complement the full research fits; a smoke run alone does not establish convergence. This report makes no hosted-CI, deployment, publication, independent-data, or clinical-validation claim.